

Research leader working on capable, verifiable AI: the mathematical capabilities of large language models across the full training stack, and verification for AI systems and agents. Working on machine learning for formal mathematics and theorem proving since 2018, from early benchmarks (CoqGym) to widely used open-source platforms (LeanDojo) and open state-of-the-art provers (Goedel-Prover).

EXPERIENCE

Apodex, Redwood City, CA Mar 2026 – Present
Lead Scientist

- Lead a research team working on (1) the mathematical capabilities of LLMs across the full stack — pre-training, mid-training, post-training, and agents — and (2) verification: formal verification for LLM training infrastructure, and verifiers that make agents efficient and reliable on long-horizon tasks.
- Set the lab's research agenda and direct external academic collaborations on theorem proving and verified code generation.

Meta Fundamental AI Research (FAIR), New York, NY Jun 2024 – Dec 2025
Research Scientist

- Co-authored *Formal Mathematical Reasoning: A New Frontier in AI* (ICML 2025 position paper, spotlight; also in *Communications of the ACM*), which lays out the research agenda connecting LLMs with formal methods.
- Advised the Goedel-Prover series (COLM 2025, ICLR 2026) — open-source theorem provers that set the state of the art among open models — and ProofOptimizer (ICLR 2026), a model that simplifies machine-generated proofs without human demonstrations.
- Collaborations on verified code generation (Verina, ICLR 2026) and neuro-symbolic olympiad problem solving (ICLR 2025, CAV 2025).

California Institute of Technology, Pasadena, CA Sep 2022 – May 2024
Computing, Data, and Society Postdoctoral Fellow

- Built [LeanDojo](#) (NeurIPS 2023, oral), an open-source platform for LLM-based theorem proving in Lean, and Lean Copilot, which runs LLM inference natively inside the Lean proof assistant; both are in regular use in the field.
- Co-organized the MATH-AI workshop and the tutorial on machine learning for theorem proving at NeurIPS 2023.

EDUCATION

Princeton University, Princeton, NJ 2022
PhD in Computer Science, advised by Jia Deng

University of Michigan, Ann Arbor, MI 2018
MS in Computer Science and Engineering

Tsinghua University, Beijing, China 2016
B.Eng. in Computer Science and B.S. in Mathematics

SELECTED PUBLICATIONS

Full list on [Google Scholar](#) (30+ papers; NeurIPS, ICML, ICLR, COLM, CACM, EMNLP, CVPR, ICCV, ECCV, CAV).

- **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
K. Yang, A. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, A. Anandkumar. NeurIPS 2023 (oral).
- **Formal Mathematical Reasoning: A New Frontier in AI**
K. Yang, G. Poesia, J. He, W. Li, K. Lauter, S. Chaudhuri, D. Song. ICML 2025 (spotlight); *Communications of the ACM*.
- **Goedel-Prover: A Frontier Model for Open-Source Automated Theorem Proving**
Y. Lin*, S. Tang*, B. Lyu, J. Wu, H. Lin, K. Yang, J. Li, M. Xia, D. Chen, S. Arora, C. Jin. COLM 2025.
- **Goedel-Prover-V2: Scaling Formal Theorem Proving with Scaffolded Data Synthesis and Self-Correction**
Y. Lin*, S. Tang*, B. Lyu*, Z. Yang*, et al. ICLR 2026.

- **Verina: Benchmarking Verifiable Code Generation**
Z. Ye, Z. Yan, T. Kasriel, J. He, K. Yang, D. Song. ICLR 2026.
- **Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean**
P. Song, K. Yang, A. Anandkumar. NeuS 2025.
- **Proving Olympiad Inequalities by Synergizing LLMs and Symbolic Reasoning**
Z. Li*, Z. Li*, W. Tang, X. Zhang, Y. Yao, X. Si, F. Yang, K. Yang[†], X. Ma[†]. ICLR 2025.
- **Generating Natural Language Proofs with Verifier-Guided Search**
K. Yang, J. Deng, D. Chen. EMNLP 2022 (oral).
- **Learning to Prove Theorems via Interacting with Proof Assistants**
K. Yang, J. Deng. ICML 2019.

OPEN SOURCE

Core developer of [LeanDojo](#), [ReProver](#), [Lean Copilot](#), and [CoqGym](#) — open-source tooling for machine learning with the Lean and Coq proof assistants.

COMMUNITY LEADERSHIP

- Co-organizer, Workshop on Mathematical Reasoning and AI (MATH-AI) at NeurIPS 2023, 2025, and 2026; co-organizer of the NeurIPS 2023 tutorial on machine learning for theorem proving.
- Area Chair: ICML 2025 & 2026, NeurIPS 2026, ECCV 2024. Reviewer for NSF panels and ERC Advanced Grants.
- Guest co-instructor, *Advanced Large Language Model Agents* (UC Berkeley & MOOC, 2025); guest lecturer at Caltech and CUHK.
- Invited talks at OpenAI, Google, Meta FAIR, the Simons Institute, UC Berkeley, Stanford, CMU, and other universities.
- Mentored 8+ students and junior researchers, now in PhD programs (Toronto, Cornell, CUHK, UIUC) and at industry labs.

RECOGNITION

- Siebel Scholar (2022) — 42 computer science graduate students awarded annually worldwide.
- Oral presentations at NeurIPS 2023 and EMNLP 2022; ICML 2025 spotlight; NeurIPS 2020 spotlight.
- Outstanding Reviewer, CVPR 2020 & 2021 (top 20%).
- Research covered by Scientific American and Wired.

* equal contribution [†] equal advising